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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: MBATech/BTech

Year: II Semester: III

Stream/s : Data Science

Subject: Managing Uncertainty

Time: 03 hrs (10 am to 1 pm)

Date: 25 / 11 / 2023

No. of Pages: 05

Marks: 100

Final Examination/Re-Examination (2022-2023)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

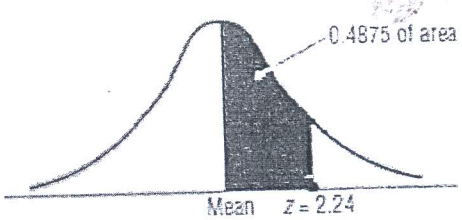
1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1 CO- 1; SO- 1; BL-2	a.	Explain classification. Also, enlist various types of classification.	[5]
CO- 1; SO-1 ; BL-1	b.	Explain the scatter plot with a suitable example.	[5]
CO-2 ; SO-2 ; BL-3	c.	A recent study of how IT professionals spend their leisure time surveyed workers employed for more than 5 years. They determined the probability of an employee having 2 weeks of vacation time to be 0.45, 1 week of vacation time to be 0.10, and 3 or more weeks of vacation time to be 0.20. Suppose 20 workers are selected at random. Answer the following questions. a) What is the probability that 8 have 2 weeks of vacation time? b) What is the probability that at most 2 of the workers have 3 or more weeks of vacation time?	[5]

<table>
<tr><td>CO- 3;
SO- 2;
BL-2</td><td>d.</td><td>For the given observations for random variables X and Y, calculate values for the following transformations.
E(X)=10, V(X)=3, E(Y)=20, V(Y)=2
a) E(3Y-7)
b) E(X-Y)
c) V(X-Y)
d) V(5X+9)</td><td>[5]</td></tr>
<tr><td>Q2
CO-1;
SO-2;
BL-5</td><td>a.</td><td>In Tarun's school, there are 25 teachers. Each teacher travels to school every morning in his or her own car. The distribution of the driving times (in minutes) from home to school for the teachers is shown in the table below:
Driving time (in minutes) Number of Teachers
0-10 3
10-20 10
20-30 6
30-40 4
40-50 2
Calculate the mean of the driving times.</td><td>[10]</td></tr>
<tr><td>CO-2 ;
SO-2 ;
BL-5</td><td>b.</td><td>Calculate the correlation coefficient of the given data
<table><tr><td>x</td><td>12</td><td>15</td><td>18</td><td>21</td><td>27</td></tr><tr><td>y</td><td>2</td><td>4</td><td>6</td><td>8</td><td>12</td></tr></table>Comment and explain the type of relationship that variables x and y exhibit from your answers.</td><td>[10]</td></tr>
<tr><td>Q3
CO-2;
SO-2;
BL-3</td><td>a.</td><td>The following data of the daily production rate of fibreglass boats from Hydrosport, Ltd. a Miami manufacturer,
17 21 18 27 17 21 20 22 18 23
The company production manager feels that a standard deviation of more than three boats a day indicates unacceptable production-rate variations. Should she be concerned about plant production rates?</td><td>[10]</td></tr>
</table>

CO-2 ; SO- 6; BL-3	b.	A bag contains 3 red and 7 black balls. Two balls are drawn at random without replacement. If the second ball is red, what is the probability that the first ball is also red?	[10]
Q4 CO-2; SO-6; BL-2	a.	Three urns are there containing white and black balls; the first urn has 3 white and 2 black balls; the second urn has 2 white and 3 black balls and the third urn has 4 white and 1 black balls. Without any biasing, one urn was chosen, and from that one ball was chosen randomly which was white. What is the probability that it came from the third urn?	[5]
CO-3; SO-6; BL-3	b.	On average, a pair of running shoes can last 18 months. The length of time running shoes last is exponentially distributed. What is the probability that a pair of running shoes last more than 15 months?	[5]
CO-3; SO-2; BL-3	c.	The average weight gained by a person over the winter months is uniformly distributed and ranges from 0 to 30 lbs. Find the probability that a person will gain between 10 and 15 lbs in the winter months. Also, calculate the expected value and standard deviation for the given distribution.	[10]
Q5 CO-2; SO-2; BL-2	a.	Calculate the variance for the given data x_i f_i 10 1 4 3 6 5 8 1	[10]
CO- 3; SO- 6; BL-5	b.	If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective.	[10]
Q6 CO-3; SO-2; BL-3	a.	How the range of a function of a random variable X can be generated using the range of X. Let X be a discrete random variable with $P_X(k)=1/4$ for $k=-1,0,1,2$. Let $Y=3\|X\|$. Find the range and PMF of Y.	[10]
CO-3 ; SO- 1; BL-1	b.	Describe the concept of PMF for a discrete random variable with a suitable example. Also, give PDF for exponential distribution.	[10]

Q7 CO-3; SO-6; BL-5	a.	The average number of acres burned by forest and range fires in a large New Mexico county is 4,300 acres per year, with a standard deviation of 750 acres. The distribution of the number of acres burned is normal. What is the probability that between 2,500 and 4,200 acres will be burned in any given year?	[10]
CO- 3; SO- 6; BL-5	b.	In 1993, the Financial Accounting Standards Board (FASB) was considering a proposal to require companies to report the potential effect of employees' stock options on earnings per share (EPS). A random sample of 41 high-technology firms revealed that the new proposal would reduce EPS by an average of 13.8 percent, with a standard deviation of 18.9 percent. A random sample of 35 producers of consumer goods showed that the proposal would reduce EPS by 9.1 percent on average, with a standard deviation of 8.7 percent. On the basis of these samples, is it reasonable to conclude (at $\alpha = 0.10$) that the FASB proposal will cause a greater reduction in EPS for high-technology firms than for producers of consumer goods?	[10]



EXAMPLE: TO FIND THE AREA UNDER THE CURVE BETWEEN THE MEAN AND A POINT 2.24 STANDARD DEVIATIONS TO THE RIGHT OF THE MEAN, LOOK UP THE VALUE OPPOSITE 2.2 AND UNDER 0.04 IN THE TABLE; 0.4875 OF THE AREA UNDER THE CURVE LIES BETWEEN THE MEAN AND A z VALUE OF 2.24.

APPENDIX TABLE 1 AREAS UNDER THE STANDARD NORMAL PROBABILITY DISTRIBUTION BETWEEN THE MEAN AND POSITIVE VALUES OF z

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898.	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990